

Unit V: Fourier Transforms

Class 1:

1. Find the Fourier transform of the function $f(t)$ defined as

$$f(t) = \begin{cases} t & |t| \leq \alpha \\ 0 & |t| > \alpha \end{cases} \quad \text{Ans: } \frac{2i}{\omega^2} (\omega \cos \omega \alpha - \sin \omega \alpha)$$

2. Find the Fourier transform of the function $f(t)$ defined as

$$f(t) = \begin{cases} \frac{1}{2a} & |t| \leq a \\ 0 & |t| > a \end{cases} \quad \text{Ans: } \frac{\sin \omega a}{\omega a}$$
